Prediction

Adaptive Systems Programming

Data types

Numerical, categorical

Numerical data

- Quantitative
- Mathematical operators can be applied over these data (e.g., calculate the average).
- Sub-types
  - Integer data
  - Continuous data
- Examples: temperature, mass, number of likes, length, width

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Categorical data

- Values that represent categories or types
- Could be numbers (ej. 0, 1, 2), but these are **symbolic** (e.g. we cannot add or subtract them)
- Example: student id, eye color (brown, blue, black, green, etc.), character type (hero, villain), hardness of a material
- Sub-types
  - Nominal (order does not matter)
  - Ordinal (order *does* matter)

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Example

Passenger Id	Traveling class	Name	Sex	Age	Fee	Survivor?
1	3	Owen Braund	M	22	7.25	0
2	1	John Cumings	M	38	71.28	1
3	3	Laina Heikkinen	F	26	7.92	1

Fragment of the *Titanic* data set

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Prediction

Classification, regression

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What is it?

- Consists of assigning **label** to an **instance**.
- This instance is usually composed of a series of features.
- The label could be of different types.

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instance

label

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Prediction problems

- **Classification**
  - Consists of assigning the instance to a *class* (categorical label), given its features and a predefined set of *n* classes.
- **Regression**
  - Consists of assigning the instance to a real value (numerical label), given its features.

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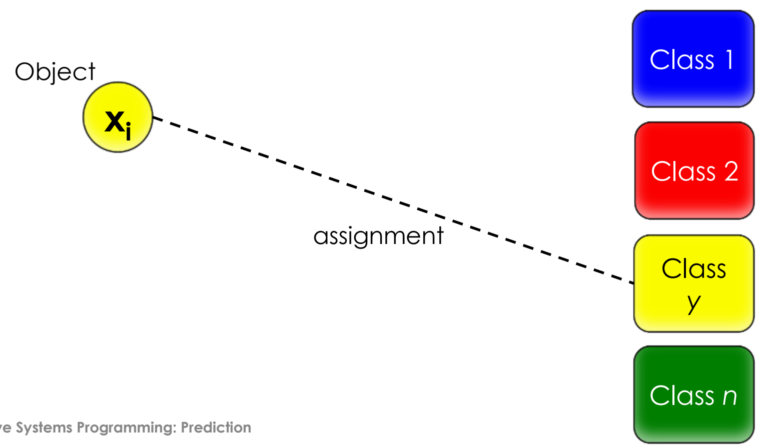
Classification

- Consists of assigning instances to **classes or** pre-defined **groups**.
- In contrast to *clustering*, the classes are already known beforehand.

Classification

- Also known as **categorization**.



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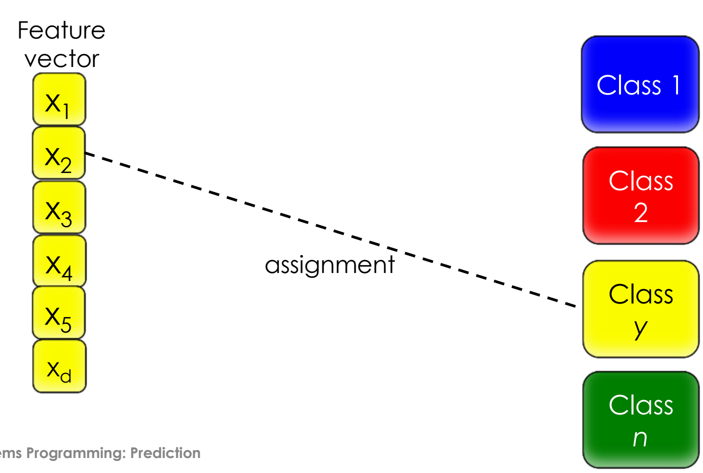

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Examples

Context: Classify a user as influencer

Comments	Likes	Posts	Views
10	5	100	30
1,000	20,000	500	2,000
200,000	100,000	1,000	50,000

Possible classes

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Classification types

- **Binary**: Two possible classes
  - It is, in general easier than having multiple classes.
- **Multi-class**: *n* possible classes
  - It is easier to address with deep learning.

Example: Sentiment analysis

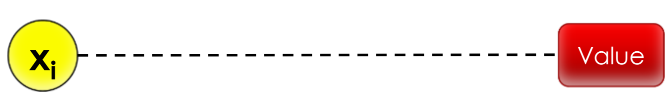
- A recent application is document sentiment analysis (opinions).
  - Three classes: Positive, negative, neutral.

  - **Positive example**: "The movie was fun."
  - **Negative example**: "The device is not functional."
  - **Neutral example**: "The motor has two sensors."

Regression

- The instance is not assigned to a class, but to a **real number**.

- Mapping of the feature vector to a real value.



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Example

Context: Determine air quality for day X

CO	PT08.S1	NMHC	C6H6	Value
2.6	1360	150	11.9	0.75
2	1292	112	9.4	0.72
2	1402	88	9.0	0.75

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References

- Tan et al. Introduction to Data Mining. Addison Wesley, EUA, 2006.